

Your Team Name

TECHNICAL APPROACH

Technology Stack | System Architecture | Implementation | Working Prototype

SMART INDIA HACKATHON
2026
SIH 2026 OFFICIAL

TECH STACK

Modern | Secure | Scalable

Frontend

- React.js
- Vite
- Tailwind CSS
- TypeScript

Web3 Integration

- Wagmi v2
- Viem
- MetaMask

Smart Contracts

- Solidity ^0.8.20
- OpenZeppelin
- Foundry

Blockchain

- Sepolia (Proto)
- Permissioned EVM

Storage & Backend

- IPFS & Pinata
- PostgreSQL
- FastAPI

Security

- AES-256-GCM
- Keccak-256
- Cryptographic Sigs

SYSTEM ARCHITECTURE

Decentralized
Modular
Interoperable

Users / Orgs

- Admin
- Manager
- Auditor
- User

Frontend

React + Vite
Tailwind CSS

Backend

FastAPI (APIs)
PostgreSQL
Auth & Cache

Blockchain (EVM)

Smart Contracts (Six Modules)

IdentityRegistry

RolesAndPerms

AssetNFT

CredentialVault

AssetBridge

AuditLog

Sepolia Proto | Permissioned EVM

Storage

IPFS +
Pinata

Encrypted
Off-Chain
Payloads

IMPLEMENTATION METHODOLOGY

- 1 Identity Setup:** User/Org creates blockchain-linked identity
- 2 Role & Permission:** Assign roles (Admin | Manager | Auditor | User)
- 3 Asset Registration:** Register digital/physical assets (docs, devices)
- 4 NFT Minting / Linking:** Mint NFT for asset and link credentials
- 5 Assign / Transfer:** Secure assignment and ownership transfer
- 6 Blockchain Audit:** All actions recorded immutably on-chain
- 7 Verification:** Independent & real-time verification by authorized parties

1. Dashboard

[Live Overview]

- 24 Verified Assets
- 100% On-Chain Audit
- Active Role: Admin

Overview of assets, credentials and activity

2. Asset Details

[NFT Token #1042]

- Aaptap - Dell XPS 13
- Owner: Finance Dept
- Status: Minted & Verified

View NFT asset information & metadata

3. Credential Verification

[Verification Engine]

- Zero-Knowledge Proof
- Keccak-256 Validated
- Status: 100% Authentic

Upload and verify certificates/documents

4. Blockchain Explorer

[Sepolia Explorer]

- Tx: 0x8f4b...39a1
- Gas Used: 42,150
- Confirmed in Block #4812

On-chain transaction & hash logs

Scan QR
Try Demo ↗